

Innovation

“MRAM CAN KEEP DATA Even When Power Is Off!”

What is MRAM, and what makes it special? EFY's Nidhi Agarwal sat down with Sanjeev Aggarwal, President & CEO, Everspin to uncover the answers...

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How does MRAM work?

MRAM (magnetoresistive random access memory) operates by using two magnetic layers separated by an insulating layer. These layers can align in the same or opposite directions, altering the resistance and allowing the memory to store data as 0s and 1s. MRAM is fast, enabling data to be read and written in about 35 nanoseconds. It supports frequent writing without degradation. Additionally, it retains data when powered off and functions across a wide range of temperatures.

How is MRAM different from other non-volatile memories?

Compared to other memory

types, such as battery-backed SRAM, MRAM has distinct advantages. If the SRAM battery fails, all data is lost, and its read and write capabilities are limited. In contrast, Everspin MRAM is persistent and supports unlimited read and write cycles. Ferroelectric memory, another option, performs well at low voltages but can face reliability issues at high temperatures. MRAM, however, operates reliably across a broad temperature range, preserving data safely even during unexpected power failures.

How are the two types of MRAM different?

Toggle MRAM and spin-transfer torque (STT) MRAM are distinct in

their operation. In toggle MRAM, a magnetic field switches the direction of the top magnetic layer, known as the free layer, while the bottom layer remains fixed. For STT MRAM, the direction is altered by passing a current through an insulating layer between the two magnetic layers.

Toggle MRAM cannot be miniaturised extensively due to the size required to generate the magnetic field for switching. It is available in sizes ranging from 128 kilobits to 16 megabits. STT MRAM, on the other hand, can be scaled smaller with technological advancements, is available in sizes from 16 megabits to 128 megabits, and even up to 1 gigabit, making it suitable for replacing traditional RAM.